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Presentability & the six functors

Last time Proved that for M an abelian group, the Fourier transform

$$\Phi_M : \mathrm{Fun}(M, \mathrm{Sp}) \longrightarrow \mathrm{QCoh}(BD_M)$$

is an equivalence. Also showed that for a cone $\sigma \subset N_{\mathbb{R}}$,

$$\Phi_\sigma : \mathrm{Fun}(\Theta(\sigma)^{op}, \mathrm{Sp}) \longrightarrow \mathrm{QCoh}(X_\sigma / T)$$

is an equivalence.

Today Start constructing (compatible) lax sym. mon functors

$$\mathrm{Fun}(\Theta(\sigma)^{op}, \mathrm{Sp}) \longrightarrow \mathrm{Sh}(M_{\mathbb{R}}, \mathrm{Sp})$$

These will land in modules over an idempotent algebra $\omega_{\sigma^\vee}$, and be sym mon with this target.

- To do so, we need the six-functor formalism so we'll start by reviewing that.

- It'll be convenient to understand a bit about **presentable ∞-categories**.

Presentable ∞-categories

Idea Want a well-behaved class of large ∞-categories with colimits that are controlled by a small amount of data

Def An infinite cardinal κ is **regular** if no set of cardinality κ is the union of fewer than κ sets of cardinality $<\kappa$

Ex The smallest infinite cardinal $\omega = \aleph_0$ is regular

Def Let κ be a regular cardinal. An ∞-category I is **κ-filtered** if for every κ-small ∞-category K, every diagram $K \to I$ admits an extension $K^{\triangleright} \longrightarrow I$
If $\kappa = \omega$, we usually just say **filtered** instead of ω-filtered

Note When $\kappa = \omega$, this just says every finite diagram has a cone under it.
Exercise. Check that this agrees with the def of filteredness in 1-category theory.

Def Let κ be a regular cardinal and let $\mathcal{C}$ be an ∞-category with κ-filtered colimits. We say that an object $X \in \mathcal{C}$ is **κ-compact** if

$$\mathrm{Map}_{\mathcal{C}}(X,-) : \mathcal{C} \longrightarrow \mathrm{An}$$

preserves κ-filtered colimits. Write $\mathcal{C}^{\kappa} \subset \mathcal{C}$ for the full subcat spanned by the κ-compact objects.
We usually use the term **compact** instead of ω-compact.
Def. Let κ be a regular cardinal and let $\mathcal{C}$ and $\mathcal{D}$ be ∞-cats with κ-filtered colimits. A functor $F : \mathcal{C} \longrightarrow \mathcal{D}$ is **κ-finitary** if F preserves κ-filtered colimits.

Def Let $\mathcal{C}_0$ be a small ∞-category and let κ be a regular cardinal. The **Ind_{κ}-completion** of $\mathcal{C}_0$ is the universal ∞-cat equipped with a functor $y : \mathcal{C}_0 \longrightarrow \mathrm{Ind}_{\kappa}(\mathcal{C}_0)$ satisfying the following properties:

(1) The ∞-category $\mathrm{Ind}_{\kappa}(\mathcal{C}_0)$ has κ-filtered colimits.

(2) For any ∞-category $\mathcal{D}$ with κ-filtered colimits, precomp with y defines an equivalence

$$\mathrm{Fun}^{\kappa\text{-fin}}(\mathrm{Ind}_\kappa(\mathcal{C}_0), \mathcal{D}) \xrightarrow{\sim} \mathrm{Fun}(\mathcal{C}_0, \mathcal{D}).$$

Construction $\mathrm{Ind}_\kappa(\mathcal{C}_0) \subset \mathrm{PSh}(\mathcal{C}_0)$ is the smallest full subcategory of $\mathcal{C}$ containing the image of the Yoneda embedding and closed under κ-filtered colimits, and y is the Yoneda embedding

Def Let $\mathcal{C}$ be an ∞-category with κ-filtered colimits. We say that $\mathcal{C}$ is **κ-compactly generated** if the unique κ-finitary extension of the inclusion $\mathrm{Ind}_\kappa(\mathcal{C}^\kappa) \longrightarrow \mathcal{C}$ is an equivalence

An ∞-category $\mathcal{C}$ is **accessible** if $\mathcal{C}$ is κ-compactly gen for some reg cardinal κ. A functor between accessible ∞-cats is **accessible** if it is κ-finitary for some regular cardinal κ.

Ex An is compactly generated. For an $\mathbb{E}_\infty$-ring A, the ∞-cat Mod_A is compactly generated.

Ex If X is a locally compact Hausdorff space, then $\mathrm{Sh}(X, \mathrm{An})$ is $\aleph_1$-compactly generated. An observation of Neeman is that if X is a non-compact connected manifold, then the only compact object of $\mathrm{Sh}(X; \mathrm{An})$ is the initial object. In particular, $\mathrm{Sh}(X; \mathrm{An})$ is not compactly generated.

Def An ∞-category $\mathcal{C}$ is **presentable** if $\mathcal{C}$ is accessible and has all colimits.

Facts about presentable ∞-categories

Thm A presentable ∞-category has all limits

Adjoint Functor Thm. Let $F: \mathcal{C} \longrightarrow \mathcal{D}$ be a functor between presentable ∞-categories. Then:

(1) F is a left adjoint if and only if F preserves colimits.

(2) F is a right adjoint if and only if F is accessible and preserves limits.

Ntn We write $\mathrm{Pr}^L \subset \mathrm{Cat}_\infty$ for the ∞-category of presentable ∞-categories and **left adjoints**. Similarly, $\mathrm{Pr}^R \subset \mathrm{Cat}_\infty$ denotes the subcategory of presentable ∞-categories and **right adjoints**. Note that

$$(\mathrm{Pr}^L)^{op} \simeq \mathrm{Pr}^R$$

Thm. The ∞-categories Pr^L and Pr^R preserve limits and the forgetful functors $\mathrm{Pr}^L \longrightarrow \mathrm{Cat}_\infty$ and $\mathrm{Pr}^R \longrightarrow \mathrm{Cat}_\infty$ preserve them.

Ex. If X is a stack that can be written as a small colimit of representables, then $QCoh(X)$ is presentable

The tensor product

Def Let $\mathcal{C}, \mathcal{D} \in \mathrm{Pr}^L$. The **tensor product** $\mathcal{C} \otimes \mathcal{D}$ is the universal presentable ∞-category equipped with a functor $\otimes : \mathcal{C} \times \mathcal{D} \to \mathcal{C} \otimes \mathcal{D}$ satisfying

(1) $\otimes : \mathcal{C} \times \mathcal{D} \longrightarrow \mathcal{C} \otimes \mathcal{D}$ preserves colimits separately in each variable

(2) If $\mathcal{E} \in \mathrm{Pr}^L$, then restriction along $\otimes$ defines an equivalence

$$\mathrm{Fun}^{\mathrm{colim}}(\mathcal{C} \otimes \mathcal{D}, \mathcal{E}) \xrightarrow{\sim} \mathrm{Fun}^{\mathrm{colim},\mathrm{colim}}(\mathcal{C} \times \mathcal{D}, \mathcal{E})$$

colimit-preserving

preserves colims sep. in each variable

Fact The tensor product refines to a sym mon structure on Pr^L with unit An

Construction For $\mathcal{C}$ and $\mathcal{D}$ presentable, we have

$$\mathcal{C} \otimes \mathcal{D} \simeq \mathrm{Fun}^{\lim}(\mathcal{C}^{op}, \mathcal{D}) \simeq \mathrm{Fun}^{\lim}(\mathcal{D}^{op}, \mathcal{C})$$

This makes the functionality in **right** adjoints clear:

$$\mathcal{C}\otimes(-) \simeq \mathrm{Fun}^{\lim}(\mathcal{C}^{op}, -) : \mathrm{Pr}^R \longrightarrow \mathrm{Pr}^R$$

is just given by post-composition.

Ex If $i: \mathcal{D} \hookrightarrow \mathcal{E}$ is a fully faithful right adjoint of presentable ∞-categories, then $\mathcal{C}\otimes i : \mathcal{C}\otimes\mathcal{D} \longrightarrow \mathcal{C}\otimes\mathcal{E}$ is fully faithful.

Ex If A and B are E_∞-rings, then

$$\mathrm{Mod}_A \otimes \mathrm{Mod}_B \simeq \mathrm{Mod}_{A\otimes B}$$

Ex If $\mathcal{C}_0$ is a small ∞-category,

$$\mathrm{Fun}(\mathcal{C}_0, \mathrm{An}) \otimes \mathcal{E} \simeq \mathrm{Fun}(\mathcal{C}_0, \mathcal{E})$$

Observation The ∞-category $\mathrm{CAlg}(\mathrm{Pr}^L)$ is equivalent to the subcategory of $\mathrm{CMon}(\mathrm{Cat}_\infty)$ with objects presentable ∞-cats $\mathcal{C}$ with a sym. mon. structure $\otimes : \mathcal{C}\times\mathcal{C} \longrightarrow \mathcal{C}$ that preserves colims sep. in each variable. Morphisms are symmetric monoidal colimit-preserving functors

Objects of $\mathrm{CAlg}(\mathrm{Pr}^L)$ are called **presentably symmetric monoidal ∞-categories.**

Ex. Both $(\mathrm{An}, \times)$ and $(\mathrm{Sp}, \otimes)$ are idempotent algebras in Pr^L

Sheaves and $*$-functoriality

Proposition Let X be a top space and $\mathcal{E}$ a presentable ∞-category. Then the inclusion

$$\mathrm{Sh}(X;\mathcal{E}) \hookrightarrow \mathrm{Fun}(\mathrm{Open}(X)^{op},\mathcal{E})$$

admits a left adjoint L called **Sheafification**. Moreover

(1) $\mathrm{Sh}(X;\mathcal{E})$ is presentable

(2) If $\mathcal{E}$ is compactly generated, L is left exact

(3) There is a natural equivalence $\mathrm{Sh}(X;\mathrm{An})\otimes\mathcal{E} \xrightarrow{\sim} \mathrm{Sh}(X;\mathcal{E})$ compatible with $\mathrm{Fun}(\mathrm{Open}(X)^{op},\mathrm{An})\otimes\mathcal{E} \xrightarrow{\sim} \mathrm{Fun}(\mathrm{Open}(X)^{op},\mathcal{E})$

Notation Let $f: X \longrightarrow Y$ be a map of top spaces. We write

$$f_*: \mathrm{PSh}(X;\mathrm{An}) \longrightarrow \mathrm{PSh}(Y;\mathrm{An})$$

for the functor $f_*(F)(V) := F(f^{-1}(V))$.

- Note that f_* carries sheaves to sheaves

- Note that f_* preserves limits and κ-filtered colimits for any κ such that the inclusions $Sh(X;An) \subset PSh(X;An)$ and $Sh(Y;An) \subset PSh(Y;An)$ do

- So f_* admits a left adjoint f^*. Moreover, for $V \subset Y$ open,

$$f^*(\mathcal{F}(V)) \simeq \mathcal{F}(f^{-1}(V))$$

Consequence For $\mathcal{E}$ any presentable ∞-category, by tensoring with $\mathcal{E}$ we obtain an adjunction

$$f^* : Sh(Y;\mathcal{E}) \rightleftarrows Sh(X;\mathcal{E}) : f_*$$

Moreover, f_* is given by the same formula as before

Observation Let $j: U \hookrightarrow X$ be an open immersion. Then the functor $Sh(X;\mathcal{E}) \longrightarrow Sh(U;\mathcal{E})$ given by

$$F \longmapsto [V \subset U \mapsto F(V)]$$

is left adjoint to j_*. Moreover, this functor preserves limits. Hence j^* admits an additional left adjoint $j_! : Sh(U, \mathcal{E}) \longrightarrow Sh(X, \mathcal{E})$.

Observe By the above, $j^* j_* \simeq id$. Hence j_* is fully faithful. As a consequence, so is $j_!$.

Recollement

Prop. Let X be a space, $i: Z \hookrightarrow X$ a closed subspace, and $j: U = X \setminus Z \hookrightarrow X$ its open complement. Then:

(1) $i_*: Sh(Z, An) \longrightarrow Sh(X, An)$ is fully faithful.

(2) $i^*: Sh(X, An) \longrightarrow Sh(Z; An)$ and $j^*: Sh(X, An) \longrightarrow Sh(U, An)$ are jointly conservative.

Write $Glue(Z, U)$ for the ∞-category of triples

$$(F_Z \in Sh(Z), F_U \in Sh(U), \phi: F_Z \longrightarrow i^* j_*(F_U)).$$

More precisely, $Glue(Z, U)$ is the pullback

$$\begin{tikzcd} Glue(Z,U) \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & Fun([1], Sh(Z; An)) \arrow[d, "target"] \\ Sh(U; An) \arrow[r, "i^* j_*"'] & Sh(Z; An) \end{tikzcd}$$

(3) The functor

$$\begin{aligned} \mathrm{Sh}(X, \mathrm{An}) &\longrightarrow \mathrm{Glue}(Z, U) \\ F &\longmapsto \left(i^*(F),\, j^*(F),\, i^*(F) \xrightarrow{i^*(\text{unit})} i^* j_* j^*(F) \right) \end{aligned}$$

is an equivalence. The inverse sends $(F_Z, F_U, \phi\colon F_Z \longrightarrow i^* j_*(F_U))$ to the pullback of

$$i_*(F_Z) \xrightarrow{i_*(\phi)} i_* i^* j_*(F_U) \xleftarrow{\text{unit} \circ j_*} j_*(F_U).$$

Let $\mathcal{E}$ be a presentable ∞-category. Then (1) is true with An replaced by $\mathcal{E}$. If $\mathcal{E}$ is compactly generated or stable, then (2) and (3) are true with An replaced by $\mathcal{E}$.

The six-functor formalism

Def A map of locally compact Hausdorff spaces $f: X \to Y$ is **proper** if for every compact subspace $K \subset Y$, the preimage $f^{-1}(K)$ is compact.

Fact Proper maps are stable under composition and basechange.

Thm (Lurie). If $f: X \to Y$ is a proper map of locally compact Haus spaces, then

$$f_*: Sh(X, An) \longrightarrow Sh(Y, An)$$

preserves filtered colimits. As a consequence, for any presentable stable ∞-category $\mathcal{E}$, the functor

$$f_*: Sh(X; \mathcal{E}) \longrightarrow Sh(Y; \mathcal{E})$$

admits a right adjoint $f^!$

Ntn Write $\mathrm{LCH} \subset \mathrm{Top}$ for the full subcategory spanned by the locally compact Hausdorff spaces. The subcategory $\mathrm{LCH} \subset \mathrm{Top}$ is closed under finite limits.

Observe Every map $f : X \longrightarrow Y$ admits a factorization $f = pj$ where j is an open immersion and p is proper. For example, take

$$\begin{tikzcd}
X \arrow[r, hook, "j", "\text{open}"'] \arrow[rr, bend left, "f"] \arrow[dr, hook, bend right, "\text{open}"'] & X^+ \times_{Y^+} Y \arrow[r, "p"] \arrow[d, hook] \arrow[dr, phantom, "\lrcorner", very near start] & Y \arrow[d, hook, "\text{open}"] \\
& X^+ \arrow[r, "f^+"'] & Y^+
\end{tikzcd},$$

where $(-)^+$ denotes one-point compactification

Idea We'd like to define an adjunction

$$\begin{tikzcd}
\mathrm{Sh}(X;\mathcal{C}) \arrow[r, bend left, "f_!"] \arrow[r, phantom, "\perp"] & \mathrm{Sh}(Y;\mathcal{C}) \arrow[l, bend left, "f^!"]
\end{tikzcd}$$

by setting $f_! := p_* j_!$ and $f^! := j^* p^!$

- But it isn't obvious this is independent of the choice of factorization and can be made functorial.

- Nonetheless, it is true. We'll start with this next time